

Resiliency

High Availability:

Redundancy doesn't always mean always available:

- May need to be powered on manually

HA (High availability):

- Always on always available

May include different components working together:

- Active/active can provide scalability advantages

Higher availability always means higher cost:

- There always a contingency you could add
- Upgraded power, High quality server components etc

Server clustering:

Combine two or more servers:

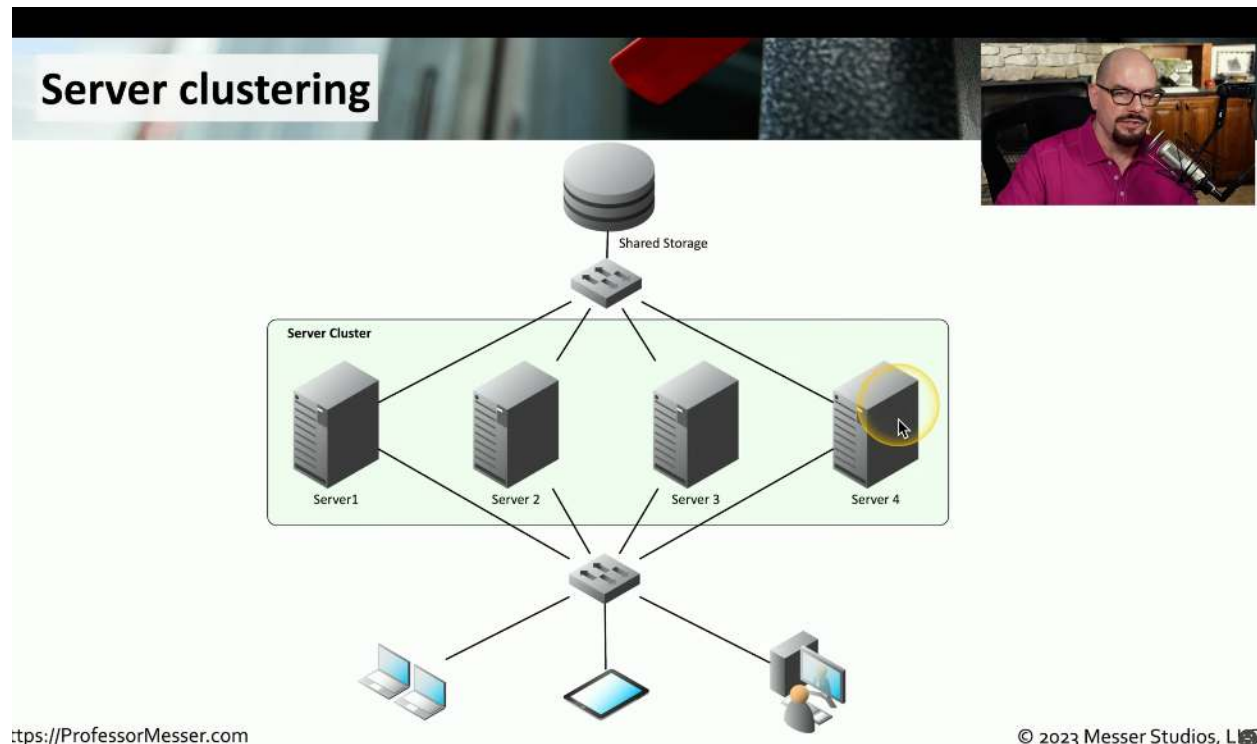
- Appears and operates as a single large server
- Users only see one device

Easily Increase capacity and availability:

- Add one server to the cluster

Usually configured in the operating system:

- All devices in the cluster are commonly use the same OS



Load Balancing

Load is distributed across multiple servers:

- these servers are often unaware of each other

Distribute the load across multiple devices:

- Can be different Operating systems

The load balancer adds or removes devices;

- Add server to increase capacity
- Remove any servers not responding

Site resiliency

Recover site is prepped:

- Data is synched

A disaster is called:

- Business processes failover to the alternate processing site

Problem Is addressed:

- This can take hours, weeks or longer

Revert back to primary location:

- Process must be document in both directions

Hot site:

An exact replica:

- Duplicate everything

Apps and software are constantly updated:

- Automated replication

Flip a switch and everything moves:

- This may be quite a few switches

Cold site:

No hardware:

- Empty building

No Data:

- Bring it with you

Warm site:

Some equipment and data but you might still need to bring stuff

Geographic dispersion:

These sites could be physically different than the organizations primary location:

- Storms, etc.

Can be a logistical challenge when trying to move stuff.

Platform diversity

Every OS contains potential security problems:

- You can't avoid them

Many security vulnerabilities are specific to a single OS:

- Windows vulnerabilities don't commonly affect Linux or macOS
- Vice versa

Use different platforms to diversify risk

Multi cloud systems

There's a lot of cloud providers

Plan for cloud outages:

- these can sometimes happen

Data is both geographically dispersed:

- A breach with one provider would not affect the other
- Plan a contingency

Continuity of operations planning (COOP)

Not everything goes according to plan:

- Disasters can cause disruption to the norm

We rely on our computer systems:

- tech is pervasive

There needs to be an alternative;

- Manual transactions
- Paper receipts etc.

Capacity planning

Match the supply to the demand:

- This isn't always an obvious equation

Too much demand:

- Application slowdowns and outages

Too much supply:

Spent too much

Requires balance

People

Some services require human intervention:

- Call center support lines

- Technology services

Too few employees:

- Recruit new staff
- It may be too time consuming to add more staff

Too many employees:

- Redeploy in other areas
- Downsize

Technology:

Pick a technology that can scale:

- Not all services can easily grow and shrink

Web services:

- Distribute the load across multiple web services

Database services:

- Cluster multiple SQL servers
- Split the database to increase capacity

Cloud services:

- Services on demand
- Seeming unlimited resources if you want to pay for them

Infrastructure

The underlying framework:

- APP servers, Network services, etc
- CPU, network storage

Physical devices:

- purchase configure and install

Cloud based devices:

- Easier to deploy
- Useful for unexpected changes

Recovery Testing

Test yourselves before an actual event:

- Scheduled update sessions

Use well defined rules of engagement;

- Do not touch production systems

Very specific scenario:

- Limited time to run the event

Evaluate response:

- Document and discuss

Tabletop exercises

Performing a full scale disaster can be costly:

- And time consuming

Many of the logistics can be determined through analysis

Get key players together for tabletop exercise

Fail over test:

A failure is often inevitable:

- Its when not if

We may be able to keep running;

- Plan for the worst

Create a redundant infrastructure:

- Multiple routers, firewalls, switches, etc.

If one stops working, fail over to operational unit.

Simulation:

Test with a simulated event;

- Phishing attack, password request, data breaches

Going phishing:

- Create a phishing email attack
- Send to your actual user community
- See who bites

Test internal security:

- Did the phishing get passed the filter

Test the users to see who clicked if the email got through

Parallel processing:

Split a process through multiple (parallel) CPUs:

- A single computer with multiple CPU cores or multiple physical CPUs
- Multiple computers

Improved performance:

- Split complex transactions across multiple processors

Improved recovery:

- Quickly identify a fault system
- Take the faulty device out of the list of available processors
- Continue operating with the remaining processors

Backups

ALWAYS HAVE A FREAKIN BACKUP.

Many different implementations:

- Total amount of data
- Type of backup
- Backup media
- Storage location
- Backup and recovery software
- Day of the week

On site backups:

- No internet link required
- Data is immediately available
- Generally less expensive than offsite

Offsite backups:

- Transfer data over the internet or WAN link
- Data is available after a disaster
- Restoration can be performed from anywhere

Organizations often use both:

- More copies of data
- More options when restoring

Frequency

How often to backup?:

- ever week day hour?

This depends on the system:

- Some systems may not change much day to day

May have multiple backup sets :

- Daily weekly and monthly

This requires significant planning:L

- Multiple backup sets across different days
- Lots of media to manage

Encryption:

History of data is on backup media:

- Some of the media maybe offsite

This makes it easy for an attacker:

- All of the data is in one place

Protect backup data using encryption:

- Everything on the backup media is unreadable
- The recovery key is required to restore the data

Especially useful for cloud backups and storage

Snapshots:

Became popular on VMs:

- Very useful for cloud environments

Take a snapshot:

- An instant backup of an entire system
- Save the current configuration and data

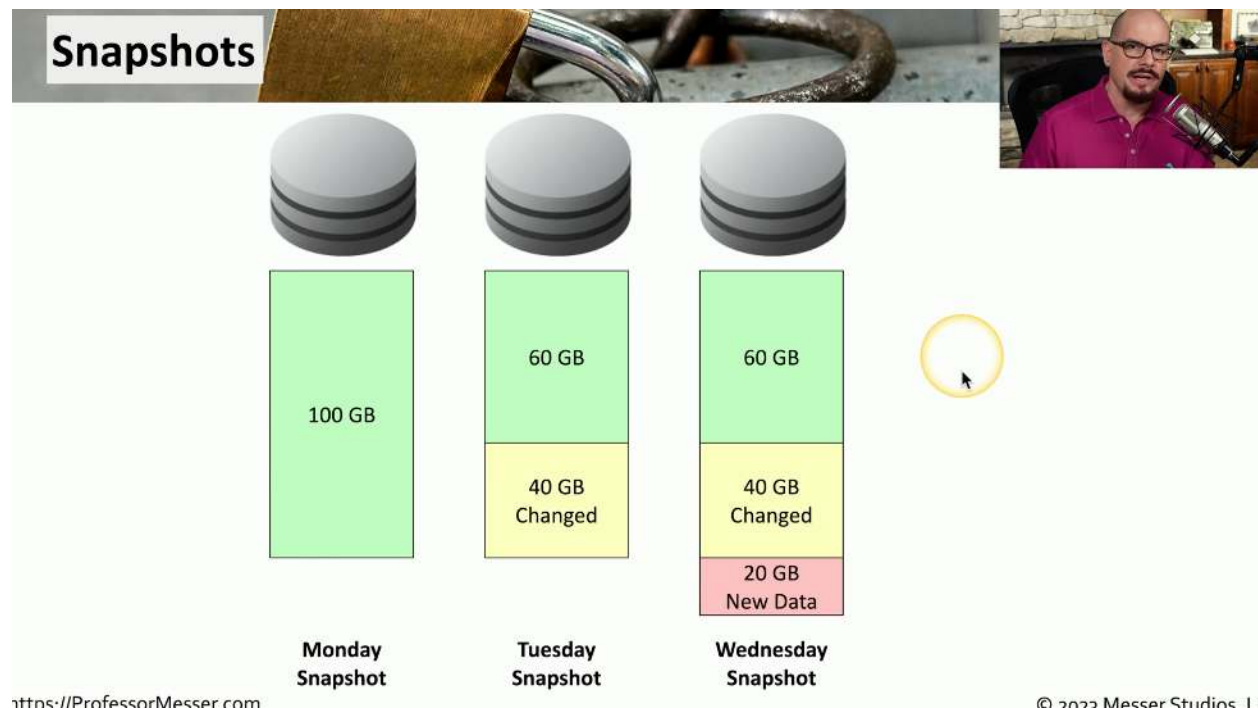
Take another snapshot after 24 hours:

- Contains only the changes between snapshots

Take a snapshot everyday:

Revert to any snapshot

Fast



TEST YOUR BACKUPS FREQUENTLY

Replication:

An ongoing almost real-time backup:

- Keep data synched in multiple locations

Data is available;

- There's always a copy

Data can be stored locally to all users:

- Replicate data to all sites

Data is recoverable all the time.

Journaling:

Power goes out will writing data to storage:

- the stored data is probably corrupted.

Recovery could be complicated;

- Remove corrupted files, restore from backup

Before writing to storage journal entry:

- After the journal is written, write the data to storage

After the data is written to storage, Update the journal:

- Clear the entry and get ready for the next

Power Resiliency

Power is the foundation of our technology:

- Its important to properly engineer and plan for outages

We usually don't make our own power;

- Power is likely provided by third parties
- We can't control power availability

There are ways to mitigate power issues:

- Short power outages
- Long term power issues

UPS

Uninterruptible power supply:

- Short term backup power
- Blackouts, brownouts, surges

UPS types:

- Offline / Standby UPS
- Line-interactive UPS
- Online/double-conversion UPS

Features:

- Auto shutdown, battery capacity, outlets, phone line suppression

Generators:

Long term power backup:

- Fuel storage required

Power an entire building:

- Some power outlets may be marked as generator powered

It may take a few minutes to get the generator up to speed:

- Use the battery UPS while the generator is starting